Predicting Critical Import Concentration in French Trade: A Machine Learning Approach to Supply-Chain Vulnerability

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***Abstract*—This paper predicts, one year in advance, which French import markets are at risk of becoming critically concentrated. The unit of observation is the HS6 product-year pair, built from 1996-2024 bilateral trade flows in the CEPII BACI database, restricted to French imports. Three classifiers (logistic regression, elastic net, random forest) are trained to flag markets whose Herfindahl-Hirschman Index (HHI) will exceed the 0.25 regulatory threshold in the following year. Under an expanding-window temporal cross-validation, all three models reach an AUC-ROC of about 0.966, driven by the strong persistence of concentration: lagged HHI values and top-exporter shares carry most of the predictive signal. On a 2022-2023 holdout, the random forest reaches an AUC of 0.970 with precision 0.651 and recall 0.529, while the logistic regression flags 93.5% of future critical markets at the cost of many false alarms. The results support an annual early-warning tool for diversification, stockpiling and trade defense policy.**

I. Problem Definition

A. Context and Motivation

The COVID-19 pandemic and the Russian invasion of Ukraine exposed structural vulnerabilities in French and European supply chains, notably the risks associated with concentrated dependencies on a limited number of supplier countries for key imports. Disruptions affecting semiconductors, active pharmaceutical ingredients, energy products, and critical raw materials highlighted that import concentration is not merely a descriptive feature of trade patterns, but a central determinant of economic resilience, price stability, and industrial continuity. These developments have intensified policy and academic interest in the notion of "strategic autonomy," with institutions such as the European Commission, CEPII, IFRI, and Institut Montaigne emphasizing the need to systematically identify and monitor "critical" import dependencies [4], particularly in the context of geopolitical tensions and the green transition.

This paper contributes to this emerging research agenda by proposing a predictive framework to identify import markets that are at risk of becoming critical. Leveraging detailed bilateral trade data over the period 1996–2024, it develops a machine learning approach to classify, one year ahead, product-level import markets likely to reach high levels of concentration. By doing so, the paper aims to provide a forward-looking tool for assessing vulnerabilities, informing public policy (e.g., diversification strategies, industrial policy, trade defense instruments), and strengthening the resilience of supply chains in an increasingly uncertain global environment.

B. Unit of Observation and Target Variable

The unit of observation is a **product** $\times$ **year** pair, where products are defined at the HS6 level (6-digit Harmonized System code). For each product-year cell, we aggregate French import flows from all source countries and compute a set of market structure indicators.

The target variable is `critical_t1`, a binary indicator equal to 1 if the Herfindahl-Hirschman Index (HHI) of the product's import market exceeds 0.25 in year $t+1$:

$$\texttt{critical_t1}_{p,t} = \mathbf{1}\left[\text{HHI}_{p,t+1} > 0.25\right] \tag{1}$$

The HHI is defined as the sum of squared market shares across all exporting countries:

$$\text{HHI}_{p,t} = \sum_i s_{i,p,t}^2, \quad s_{i,p,t} = \frac{v_{i,p,t}}{\sum_j v_{j,p,t}} \tag{2}$$

where $v_{i,p,t}$ is the value of exports from country i for product p in year t. The threshold of 0.25 corresponds to the standard definition of a highly concentrated market used by the U.S. Department of Justice and the European Commission [2], [3]. A market reaching this threshold is one where a single or very small number of exporters dominates French imports, exposing the country to supply-chain disruption risk.

C. Why This Question Matters

This prediction problem has direct relevance for public policy. A model that reliably identifies, one year in advance, which product markets are about to become critically concentrated allows trade policy administrators to take anticipatory action: engaging in supplier diversification strategies, negotiating bilateral trade agreements, building strategic stockpiles, or commissioning domestic production capacity. The horizon of one year is deliberately chosen to be operationally actionable.

More broadly, this work contributes to a growing demand for systematic, data-driven methods to monitor supply-chain

vulnerabilities across thousands of product categories simultaneously, a task intractable through traditional case-by-case analysis, and increasingly relevant in the context of the EU Critical Raw Materials Act [5] and related regulatory frameworks.

II. Data

A. Sources

The dataset is constructed from three sources:

- **BACI** (Base pour l'Analyse du Commerce International), produced by CEPII [1]. This database provides bilateral trade flows at the HS6 level for all country pairs, expressed in thousands of USD and metric tons. We use the HS96 classification covering 1996 to 2024, and restrict the sample to flows for which France (country code 251) is the importer.
- **WTO World Trade Data** [6]. Annual world merchandise trade values (in millions of USD) are used as a macroeconomic control capturing global trade conditions.
- **World Bank Pink Sheet** (CMO Historical Data) [7]. Monthly commodity prices for a selection of raw materials are aggregated to annual averages and matched to HS2 sectors. This variable was ultimately excluded from the model due to extensive missing values (see Section II-C).

B. Description of Variables

Table I presents the variables used in the analysis.

C. Data Cleaning and Missing Values

Each annual BACI file is loaded, variables are renamed, and the dataset is filtered to observations where France is the importer (country code 251). Rows with missing `quantity` values are dropped; inspection of these cases confirmed they were missing at random with no systematic pattern across products or years. The filtered annual files are stacked into a single panel and aggregated to the product $\times$ year level, yielding 120,625 product-year observations.

Feature engineering introduces lagged HHI values (up to 3 years) and a 5-year rolling average, which mechanically removes the first few years of each product's time series. The commodity price variable from the World Bank Pink Sheet was excluded: it is only available for 29 HS2 sectors out of 97 (raw materials, metals, agricultural commodities), leaving approximately 75% of observations with missing values. Retaining it after `drop_na()` would have restricted the sample to commodity-intensive sectors, severely biasing the analysis toward a non-representative subset of products. The final ML-ready dataset contains 115,519 observations covering 2000–2023.

III. Exploratory Analysis

A. Class Imbalance

The target variable `critical_t1` is highly imbalanced: only 4.2% of product-year observations (4,812 out of 115,519) correspond to a critically concentrated market. This has two important implications. First, accuracy is a misleading metric since a naive classifier predicting "non-critical" for all observations would achieve 95.8% accuracy while being entirely useless. Second, all models must be adapted to handle this imbalance; we do so through inverse frequency weighting to avoid artificially altering the training data composition.

B. HHI Distribution and Persistence

The HHI distribution is strongly right-skewed: the median is 0.081 and the mean is 0.099, reflecting that most product markets are served by a relatively diverse set of exporters. The distribution has a long right tail, with a maximum of 0.333. The 0.25 threshold sits at the 96th percentile of the distribution, capturing only the top 4% of most concentrated markets, confirming that it targets genuinely extreme concentration episodes rather than merely elevated ones.

A key empirical feature is the high autocorrelation of the HHI series. The correlation between $\text{HHI}_{p,t}$ and $\text{HHI}_{p,t-1}$ is approximately 0.83, indicating that market concentration is strongly persistent over time. This motivates the central role of lagged HHI values as predictive features and suggests that this is a favorable setting for machine learning: past concentration is a strong signal for future concentration.

C. Summary Statistics

Table II presents summary statistics for the main variables of the ML-ready dataset.

IV. Feature Engineering

Starting from the raw transaction-level BACI data, we construct a product $\times$ year panel by aggregating all import flows for a given HS6 product and year. The following transformations are applied:

a) Concentration lags: The HHI at lags 1, 2, and 3 (`hhi_lag1/2/3`) are the central predictive features, motivated by the high autocorrelation of the HHI series (approximately 0.83 at lag 1). The annual change `delta_hhi` captures short-run momentum in concentration, while the 5-year rolling average `hhi_trend_5y` captures the long-run structural trend, distinguishing temporary shocks from persistent supply-chain transitions.

b) Market structure indicators: The value share of the top exporter (`share_top_exporter`) and of the top three exporters (`share_top3_exporters`) provide complementary information to the HHI: two markets can have identical HHI but very different underlying structures. The number of distinct source countries (`Nb_exporters`) measures import diversification directly.

c) Log-transformations: Import values and unit costs span several orders of magnitude due to heterogeneity across product categories (bulk commodities, pharmaceuticals, diamonds, manufactured goods). We apply log1p transformations to `total_import` and `avg_unit_cost` to reduce skewness and limit the influence of extreme outliers.

Table I: **Variable descriptions**

Variable	Type	Description
Target		
`critical_t1`	Binary	HHI > 0.25 in year $t+1$
Concentration indicators		
`hhi_lag1/2/3`	Continuous	HHI at $t-1$, $t-2$, $t-3$
`delta_hhi`	Continuous	Annual change in HHI
`hhi_trend_5y`	Continuous	5-year rolling average of HHI
Market structure		
`share_top_exporter`	Continuous	Value share of the top exporter
`share_top3_exporters`	Continuous	Value share of top 3 exporters
`share_top_lag1`	Continuous	Top exporter share at $t-1$
`Nb_exporters`	Integer	Number of distinct source countries
Import volume		
`log_total_import`	Continuous	Log of total import value (USD)
`log_import_lag1`	Continuous	Lagged log import value
Prices		
`log_avg_unit_cost`	Continuous	Log of average unit cost (USD/ton)
Macroeconomic controls		
`crisis_year`	Binary	$= 1$ for 2008, 2009, 2020, 2021, 2022
`log_world_trade`	Continuous	Log of WTO world trade value
Sector		
`hs2`	Integer	HS2 sector code (first 2 digits of HS6)

Notes: All lagged variables are computed at the product level after sorting by year. `log_commodity_price` was excluded due to approximately 75% missing values, covering only commodity-intensive HS2 sectors; retaining it after listwise deletion would have introduced severe selection bias.

Table II: **Summary statistics, ML dataset (N = 115,519)**

Variable	Min	Mean	Median	Max	% Missing
`hhi` (current)	0.015	0.099	0.081	0.333	0%
`hhi_lag1`	0.015	0.099	0.081	0.333	0%
`delta_hhi`	−0.297	0.000	0.000	0.293	0%
`hhi_trend_5y`	0.016	0.099	0.084	0.333	4.4%
`share_top_exporter`	0.026	0.146	0.132	0.333	0%
`Nb_exporters`	1	33.6	27	185	0%
`log_total_import`	0.006	10.85	10.97	18.87	0%
`log_avg_unit_cost`	0.001	2.285	2.114	15.10	0%
`log_world_trade`	8.646	9.503	9.683	10.12	0%
`critical_t1`	0	0.042	0	1	0%

Notes: Statistics computed on the final ML-ready sample after listwise deletion. The 4.4% missing on `hhi_trend_5y` correspond to products with fewer than 5 years of observations; these rows are removed by `drop_na()`.

d) Crisis dummy: The variable `crisis_year` equals 1 for years affected by major macroeconomic shocks: 2008–2009 (global financial crisis) and 2020–2022 (COVID-19 and the Ukraine war commodity shock). These events are known to have temporarily disrupted trade patterns and may have induced transient concentration changes structurally different from ordinary year-to-year variation.

e) Sector control: The HS2 code (first two digits of the HS6 product code) captures broad sectoral heterogeneity in baseline concentration levels. For example, energy and metals systematically exhibit higher concentration than, for instance, textiles or machinery.

V. Modeling

A. Validation Strategy

A standard k-fold cross-validation would be inappropriate here because observations are structured as a time series at the product level, and features include lags of the HHI, a variable closely related to the target. Randomly mixing past and future observations into training and test sets would constitute data leakage, artificially inflating performance estimates. We therefore implement a time series split with four expanding-window folds, always training on the past and evaluating on the future:

Table III: **Time series cross-validation folds**

Fold	Training period	Test period	Train obs.
1	2000–2010	2011–2013	54,231
2	2000–2013	2014–2016	68,547
3	2000–2016	2017–2019	82,710
4	2000–2019	2020–2022	96,819

The final model is retrained on all data through 2021 (106,189 observations) and evaluated on the 2022–2023 holdout (9,330 observations).

B. Class Imbalance Handling

All three models use inverse frequency weighting to compensate for the 96/4 class imbalance. Each observation in the minority class (critical) receives a weight equal to the ratio of majority to minority class counts in the training set (approximately 23:1), while majority class observations receive weight 1. This gives equal aggregate weight to both classes without altering the training data composition through resampling.

C. Evaluation Metrics

Given the strong class imbalance, we report three complementary metrics and deliberately exclude accuracy:

- **AUC-ROC**: measures the model's ability to rank critical products above non-critical ones across all classification thresholds. This is the primary metric, as it is threshold-independent and robust to class imbalance.
- **Recall**: the share of truly critical markets correctly flagged at the 0.5 threshold. In a policy context, missing a critical market (false negative) is arguably more costly than issuing a false alarm (an assumption that could be challenged).
- **F1-score**: the harmonic mean of precision and recall, summarizing the precision-recall trade-off. A high F1 requires both few missed critical markets and few false positives.

D. Model 1: Logistic Regression (Baseline)

The baseline model is a logistic regression estimated by maximum likelihood with no regularization:

$$P(\texttt{critical_t1} = 1 \mid \mathbf{x}) = \sigma\left(\beta_0 + \sum_{j=1}^{p} \beta_j x_j\right) \qquad (3)$$

where $\sigma(\cdot)$ is the logistic sigmoid function and $\mathbf{x}$ is the vector of 15 features. This model serves as the reference against which the more complex methods are evaluated, and provides a lower bound on achievable performance under a purely linear assumption.

Table IV: **Model comparison, average performance across 4 temporal folds**

Model	AUC (mean)	AUC (sd)	Recall (mean)	F1 (mean)
Logit (Baseline)	0.966	0.007	0.935	0.432
Elastic Net	0.966	0.007	0.935	0.431
Random Forest	0.964	0.007	0.613	0.629

Notes: All models use inverse frequency weighting (ratio $\approx$ 23:1). AUC-ROC is threshold-independent and is the primary metric. Recall and F1 are computed at the 0.5 classification threshold. Standard deviations are computed across the 4 temporal folds covering test periods 2011–2022.

E. Model 2: Elastic Net

The Elastic Net penalizes the log-likelihood with a combination of L1 (Lasso) and L2 (Ridge) penalties:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left[-\ell(\boldsymbol{\beta}) + \lambda \left(\alpha |\boldsymbol{\beta}|_1 + \frac{1-\alpha}{2} |\boldsymbol{\beta}|_2^2 \right) \right] \qquad (4)$$

We set $\alpha = 0.5$, giving equal weight to the Lasso and Ridge components. The regularization parameter λ is selected by 5-fold cross-validation within each training fold, maximizing AUC. The L1 component enables automatic variable selection, while the L2 component stabilizes estimation in the presence of correlated features, notably the three HHI lags, which are highly collinear by construction.

F. Model 3: Random Forest

The Random Forest builds an ensemble of decision trees on bootstrap samples of the training data. At each node split, only a random subset of $m = \lfloor \sqrt{p} \rfloor$ features is considered, decorrelating the trees and reducing variance. Key hyperparameters:

- `ntree` $= 300$ (cross-validation) / 500 (final model)
- `mtry` $= \lfloor \sqrt{15} \rfloor = 3$
- `nodesize` $= 10$: minimum leaf size, preventing overfitting on the minority class
- `classwt`: inverse frequency weights as above

Unlike the two linear models, the Random Forest can capture non-linear interactions between features without explicit specification. For instance, a high and rising HHI combined with a very low number of exporters may signal imminent critical concentration more strongly than either feature alone.

VI. Results

A. Cross-Validation Performance

Table IV reports the average performance of each model across the four temporal folds, and Figure 1 provides a visual comparison.

In terms of AUC-ROC, all three models perform nearly identically (between 0.964 and 0.966), with overlapping standard deviations across folds. This convergence suggests that the predictive relationship between past market structure and future critical concentration is essentially linear and well-captured by the chosen features. As a consequence, the Logistic Regression already extracts almost all available signal. This interpretation is reinforced by the Elastic Net: the optimal λ

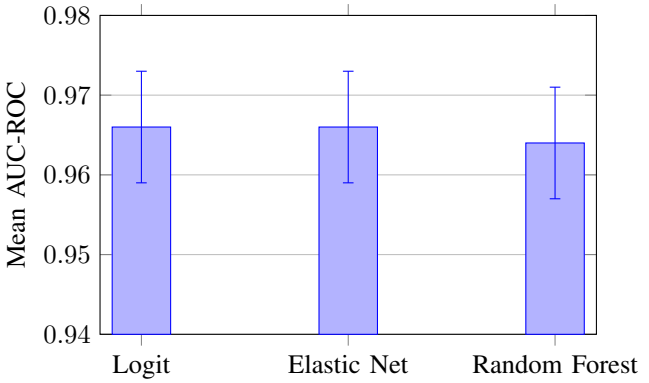


Figure 1: AUC-ROC by model, average across 4 temporal folds. Error bars denote ± 1 standard deviation across folds.

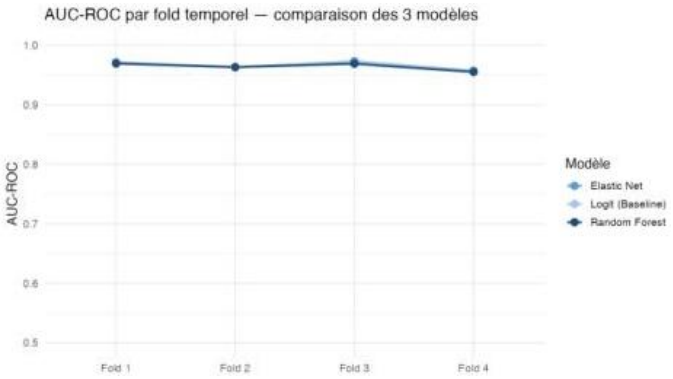


Figure 2: AUC-ROC per temporal fold for each model. The three models track each other closely across periods. The slight dip in fold 4 (test period 2020–2022) reflects the structural disruptions of the COVID-19 and Ukraine war period, which may have altered concentration dynamics in ways not fully captured by the training data.

selected by cross-validation is extremely small (approximately 7×10^{-5}), meaning the penalization is near-zero and the Elastic Net effectively converges to the unpenalized Logistic Regression.

The key differentiation emerges when moving from AUC to threshold-dependent metrics. The Logistic Regression achieves a very high recall of 0.935. This means that it correctly flags 93.5% of future critical markets, but at the cost of very low precision, as reflected in its low F1 of 0.432: it raises many false alarms for every true positive. The Random Forest achieves a more balanced trade-off: its recall drops to 0.613 but its F1 rises to 0.629, indicating substantially fewer false positives per true positive detected. Figure 2 shows that this pattern is consistent across all four temporal folds.

The choice between models therefore depends on the relative cost of each error type. If missing a critical market is considered catastrophic (because the government has no time to react once concentration materializes) then the Logistic Regression's near-perfect recall is valuable despite its high false positive rate. If, on the other hand, policy resources are limited and false alarms carry a real cost, the Random Forest's better F1 makes it the preferred tool.

Table V: **Confusion matrix, Random Forest final model (test set 2022–2023)**

	Actual	
Predicted	Non-Critical	Critical
Non-Critical	8,881	165
Critical	99	185

Notes: Test set: 9,330 observations (350 critical cases, 3.75%). AUC-ROC = 0.970. Recall = 0.529. Precision = 0.651. F1 = 0.583. Kappa = 0.569.

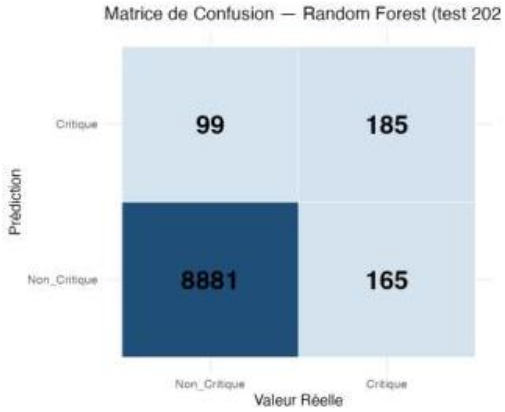


Figure 3: Confusion matrix, Random Forest (test set 2022–2023).

B. Performance on the Final Test Set (2022–2023)

The Random Forest is retrained on the full 2000–2021 period and evaluated on the 2022–2023 holdout (9,330 observations, 350 critical cases). Table V presents the confusion matrix and Figure 3 provides a visual representation.

The final model achieves an AUC of **0.970** on the holdout, slightly above the cross-validation average, in line with the stability of the estimates across folds. Among the 350 truly critical product-year cases, the model correctly identifies 185 while generating 99 false positives.

C. Feature Importance

Figure 4 displays the mean decrease in Gini impurity for each feature of the final Random Forest, measuring each variable's average contribution to reducing classification impurity across all splits and all trees.

The feature importance ranking is both economically interpretable and reassuring. The dominant predictors are the lagged HHI values (`hhi_lag1`, `hhi_lag2`, `hhi_lag3`) and the top exporter market share variables (`share_top_exporter`, `share_top_lag1`), directly reflecting the high persistence of concentration documented in Section III. The 5-year rolling average `hhi_trend_5y` also ranks highly, confirming that structural trends carry predictive information beyond the most recent level. The number

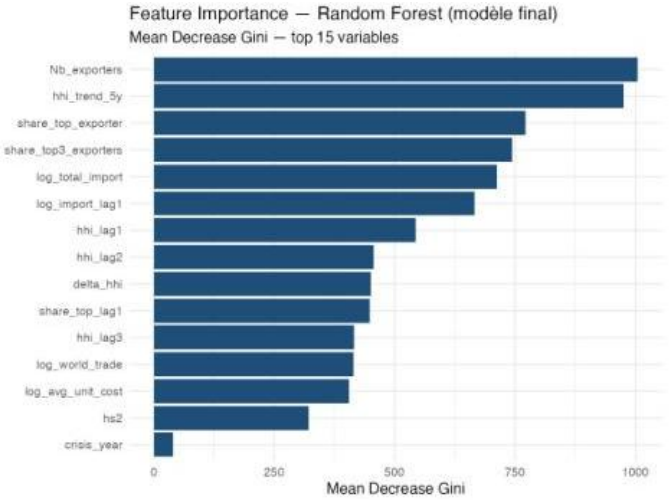


Figure 4: Feature importance, Random Forest final model. Mean Decrease Gini. Higher values indicate greater average contribution to node purity across the ensemble.

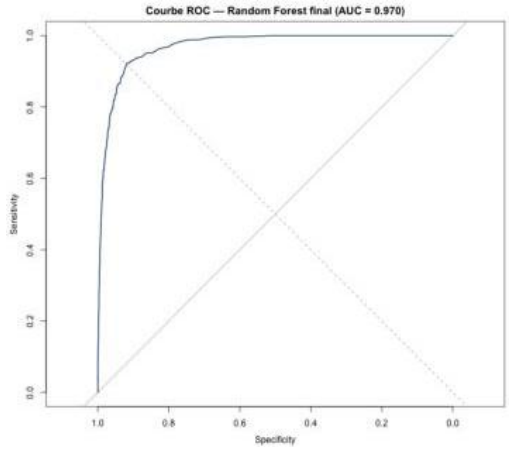


Figure 5: ROC curve, Random Forest final model (AUC = 0.970, test set 2022–2023).

of exporters (`Nb_exporters`) contributes meaningfully: markets served by few source countries are structurally more vulnerable to tipping into critical concentration. By contrast, the macroeconomic controls (`crisis_year`, `log_world_trade`) and the sector code (`hs2`) contribute less, suggesting that concentration dynamics are primarily driven by product-level idiosyncratic factors rather than aggregate macro conditions.

VII. Interpretation and Limits

A. What a Decision-Maker Can Do With This Model

This model can function as an annual early-warning system for import concentration risk. Each year, once the latest BACI trade data is available, the model generates a ranked list of products predicted to reach critical concentration in the following year, together with their predicted probability. Policy administrators can then focus on products with the highest risk scores, cross-referencing with strategic importance criteria (defense, health, energy) to prioritize intervention.

The practical choice between the Logistic Regression and the Random Forest maps to an institutional preference. A ministry of economy concerned above all with not missing critical vulnerabilities would favor the Logistic Regression's near-perfect recall. An agency with limited resources, where false alarms carry a real budgetary cost, would prefer the Random Forest's more precise predictions.

B. Limitations

Several limitations should be acknowledged. First, data availability: BACI is published with an approximate two-year lag, which means that in real deployment, the model would effectively predict concentration two to three years ahead, a harder task than the one evaluated here. Second, the HHI threshold of 0.25 is a conventional regulatory benchmark but is somewhat arbitrary, and the sensitivity of results to alternative thresholds is not tested. Third, the model is purely predictive: it identifies when concentration is likely to increase but not why, and therefore cannot directly prescribe which policy lever to pull. Adding geopolitical variables would be a natural and valuable extension, particularly given the structural shifts in global supply chains observed since 2022. Fourth, the commodity price variable was excluded due to extensive missing values, which may reduce accuracy for raw material sectors where supply shocks are most consequential. Finally, the model is trained on French import data only; its applicability to other importers would require retraining and validation on different national datasets.

VIII. Conclusion

This paper develops a machine learning framework to predict, one year in advance, whether a given product's French import market will reach a critical level of concentration. Using 25 years of bilateral trade data from BACI (1996–2024) and a rigorous time series validation strategy, we train and compare three models: a Logistic Regression baseline, an Elastic Net, and a Random Forest.

The central finding is that all three models achieve high and near-identical AUC-ROC scores of approximately 0.966, indicating that critical import concentration is highly predictable from past market structure. The near-zero regularization selected by the Elastic Net confirms that the predictive relationship is essentially linear and well-specified, making the Logistic Regression a surprisingly strong baseline.

The key differentiation between models lies in the precision-recall trade-off at the classification threshold. The Logistic Regression achieves near-perfect recall (0.935) but low precision, generating many false alarms. The Random Forest achieves a substantially better F1 score (0.629 vs. 0.432), offering a more balanced and operationally useful output for resource-constrained policy settings. Feature importance analysis confirms that the dominant predictors are lagged HHI values and top exporter market shares, validating the feature

engineering choices and providing an economically coherent picture of concentration dynamics.

This work illustrates both the potential and the limits of data-driven approaches to economic security. The model provides a scalable, annually updatable early-warning tool covering thousands of products simultaneously. Its key limitation is the absence of forward-looking geopolitical variables, which may matter most precisely for the products where early warning is most valuable.

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